

Class 13: RNA-Seq analysis mini-project

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Background

Today we will run through a complete RNASeq analysis

The data for for hands-on session comes from GEO entry: GSE37704, which is associated with the following publication:

Trapnell C, Hendrickson DG, Sauvageau M, Goff L et al. “Differential analysis of gene regulation at transcript resolution with RNA-seq”. Nat Biotechnol 2013 Jan;31(1):46-53. PMID: 23222703

The authors report on differential analysis of lung fibroblasts in response to loss of the developmental transcription factor HOXA1. Their results and others indicate that HOXA1 is required for lung fibroblast and HeLa cell cycle progression. In particular their analysis show that “loss of HOXA1 results in significant expression level changes in thousands of individual transcripts, along with isoform switching events in key regulators of the cell cycle”. For our session we have used their Sailfish gene-level estimated counts and hence are restricted to protein-coding genes only.

Data Import

Q. Complete the code below to remove the troublesome first column from count-Data

```
counts <- read.csv("GSE37704_featurecounts.csv", row.names=1)
metadata <- read.csv("GSE37704_metadata.csv")
```

Check correspondence of metadata and counts (i.e. that columns in counts match the rows in metadata).

```
metadata
```

	id	condition
1	SRR493366	control_sirna
2	SRR493367	control_sirna
3	SRR493368	control_sirna
4	SRR493369	hoxa1_kd
5	SRR493370	hoxa1_kd
6	SRR493371	hoxa1_kd

```
colnames(counts)
```

```
[1] "length"      "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370"
[7] "SRR493371"
```

```
metadata$id
```

```
[1] "SRR493366" "SRR493367" "SRR493368" "SRR493369" "SRR493370" "SRR493371"
```

Fix to remove that first length column

```
counts <- counts[,-1]
```

Also let's remove low count genes

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

```
tot.counts <- rowSums(counts)
head(tot.counts)
```

```
ENSG00000186092 ENSG00000279928 ENSG00000279457 ENSG00000278566 ENSG00000273547
                0                0                183                0                0
ENSG00000187634
                1129
```

Let's remove all zero count genes

```
zero.inds <- tot.counts == 0
head(zero.inds)
```

```
ENSG00000186092 ENSG00000279928 ENSG00000279457 ENSG00000278566 ENSG00000273547
                TRUE                TRUE                FALSE                TRUE                TRUE
ENSG00000187634
                FALSE
```

```
# no more zero counts!
counts <- counts[!zero.inds,]
```

```
test_cols <- all(colnames(counts) == metadata$id)
```

```
if ( !test_cols ){
  message("Wow.... there is a problem with the metadata counts setup")
}
```

Setup for DESeq

```
library(DESeq2)
```

Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
setequal, union

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

```
dds <- DESeqDataSetFromMatrix(countData = counts,  
                              colData = metadata,  
                              design = ~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
#tilda, ~
```

Run DESeq

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

Get Results

```
res <- results(dds)
```

Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
summary(res)
```

```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
low counts [2]    : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

```
head(res)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 6 rows and 6 columns
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01

```
      padj
      <numeric>
ENSG00000279457 6.86555e-01
ENSG00000187634 5.15718e-03
ENSG00000188976 1.76553e-35
ENSG00000187961 1.13413e-07
ENSG00000187583 9.19031e-01
ENSG00000187642 4.03379e-01
```

Add annotation

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
columns(org.Hs.eg.db)
```

[1]	"ACCNUM"	"ALIAS"	"ENSEMBL"	"ENSEMBLPROT"	"ENSEMBLTRANS"
[6]	"ENTREZID"	"ENZYME"	"EVIDENCE"	"EVIDENCEALL"	"GENENAME"
[11]	"GENETYPE"	"GO"	"GOALL"	"IPI"	"MAP"
[16]	"OMIM"	"ONTOLOGY"	"ONTOLOGYALL"	"PATH"	"PFAM"
[21]	"PMID"	"PROSITE"	"REFSEQ"	"SYMBOL"	"UCSCKG"
[26]	"UNIPROT"				

```
res$symbol = mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="SYMBOL",
                     multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez= mapIds(org.Hs.eg.db,
                   keys=row.names(res),
                   keytype="ENSEMBL", # format of our genes
                   column="ENTREZID",
                   multiVals = "first") # new format we want to add
```

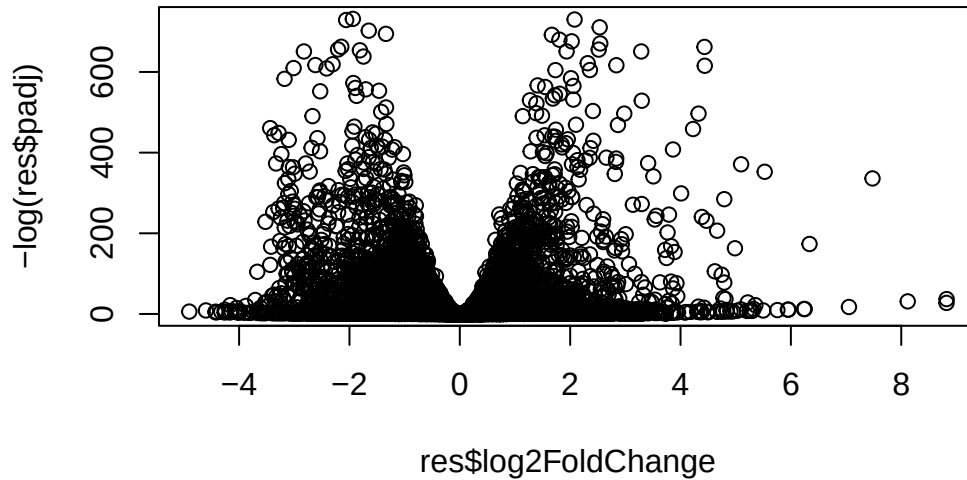
'select()' returned 1:many mapping between keys and columns

```
res$name= mapIds(org.Hs.eg.db,
                  keys=row.names(res),
                  keytype="ENSEMBL", # format of our genes
                  column="GENENAME",
                  multiVals = "first") # new format we want to add
```

'select()' returned 1:many mapping between keys and columns

Visualize Results

```
plot( res$log2FoldChange, -log(res$padj) )
```



```
library(ggplot2)
```

```
my_cols <- rep("gray", nrow(res))
```

```
# Color red the genes with absolute fold change above 2
```

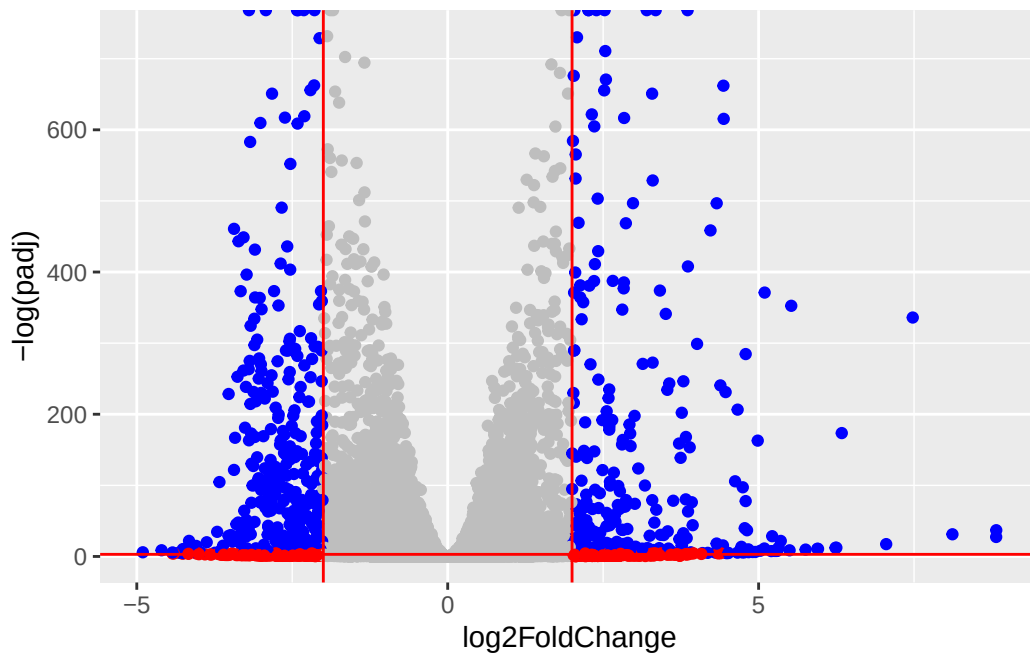
```
my_cols[ abs(res$log2FoldChange) > 2 ] <- "red"
```

```
# Color blue the genes with adjusted p-value below 0.01 and absolute fold change more than 2
```

```
my_cols[ res$padj < 0.01 & abs(res$log2FoldChange) > 2 ] <- "blue"
```

```
ggplot(res) +  
  aes(log2FoldChange, -log(padj)) +  
  geom_point(col=my_cols) +  
  geom_vline(xintercept = c(-2,2), col="red") +  
  geom_hline(yintercept = -log(0.05), col="red")
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (``geom_point()``).



Pathway analysis

Q. Use the `mapIDs()` function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
library(gage)
```

```
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
[49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"
[57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
[65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
[73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
[81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
[89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
[97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
[105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
[113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
[121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
[129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
[137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
[145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
[153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
[161] "9583" "9615"
```

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
<NA>      148398      26155      339451      84069      84808
0.17925708 0.42645712 -0.69272046 0.72975561 0.04057653 0.54281049
```

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
library(pathview)
```

```
#####  
Pathview is an open source software package distributed under GNU General  
Public License version 3 (GPLv3). Details of GPLv3 is available at  
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to  
formally cite the original Pathview paper (not just mention it) in publications  
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/elizabethgilmore/Desktop/UCSD_Salk/BGGN213/BGGN213_github/

Info: Writing image file hsa04110.pathview.png

```
# A different PDF based output of the same data
```

```
pathview(gene.data=foldchanges, pathway.id="hsa04110", kegg.native=FALSE)
```

'select()' returned 1:1 mapping between keys and columns

Warning: reconcile groups sharing member nodes!

```
      [,1] [,2]  
[1,] "9"  "300"  
[2,] "9"  "306"
```

Info: Working in directory /Users/elizabethgilmore/Desktop/UCSD_Salk/BGGN213/BGGN213_github/

Info: Writing image file hsa04110.pathview.pdf

```
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$greater)[1:5]
```

```
# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/elizabethgilmore/Desktop/UCSD_Salk/BGGN213/BGGN213_github/0
```

```
Info: Writing image file hsa04640.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/elizabethgilmore/Desktop/UCSD_Salk/BGGN213/BGGN213_github/0
```

```
Info: Writing image file hsa04630.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/elizabethgilmore/Desktop/UCSD_Salk/BGGN213/BGGN213_github/0
```

```
Info: Writing image file hsa00140.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/elizabethgilmore/Desktop/UCSD_Salk/BGGN213/BGGN213_github/0
```

```
Info: Writing image file hsa04142.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/elizabethgilmore/Desktop/UCSD_Salk/BGGN213/BGGN213_github/0
```

```
Info: Writing image file hsa04330.pathview.png
```

```
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	8.995727e-06	-4.378644	8.995727e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.246882e-03	-3.059466	1.246882e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa04114 Oocyte meiosis	3.784520e-03	-2.698128	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	8.961413e-03	-2.405398	8.961413e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001448312	121	8.995727e-06
hsa03030 DNA replication	0.007586381	36	9.424076e-05
hsa03013 RNA transport	0.066915975	144	1.246882e-03
hsa03440 Homologous recombination	0.121861535	28	3.066756e-03
hsa04114 Oocyte meiosis	0.121861535	102	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	0.212222694	53	8.961413e-03

Insert figure for this pathway

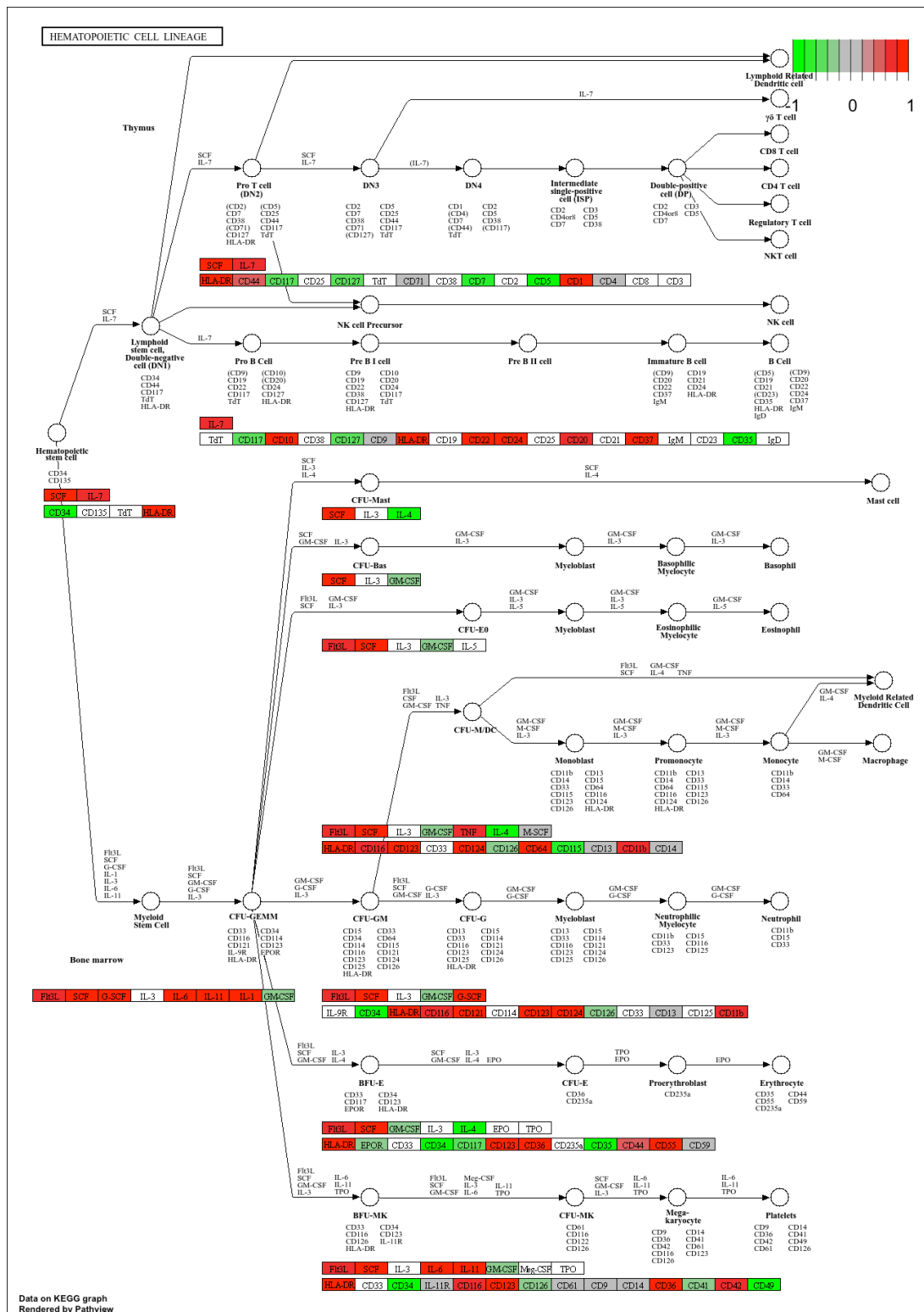
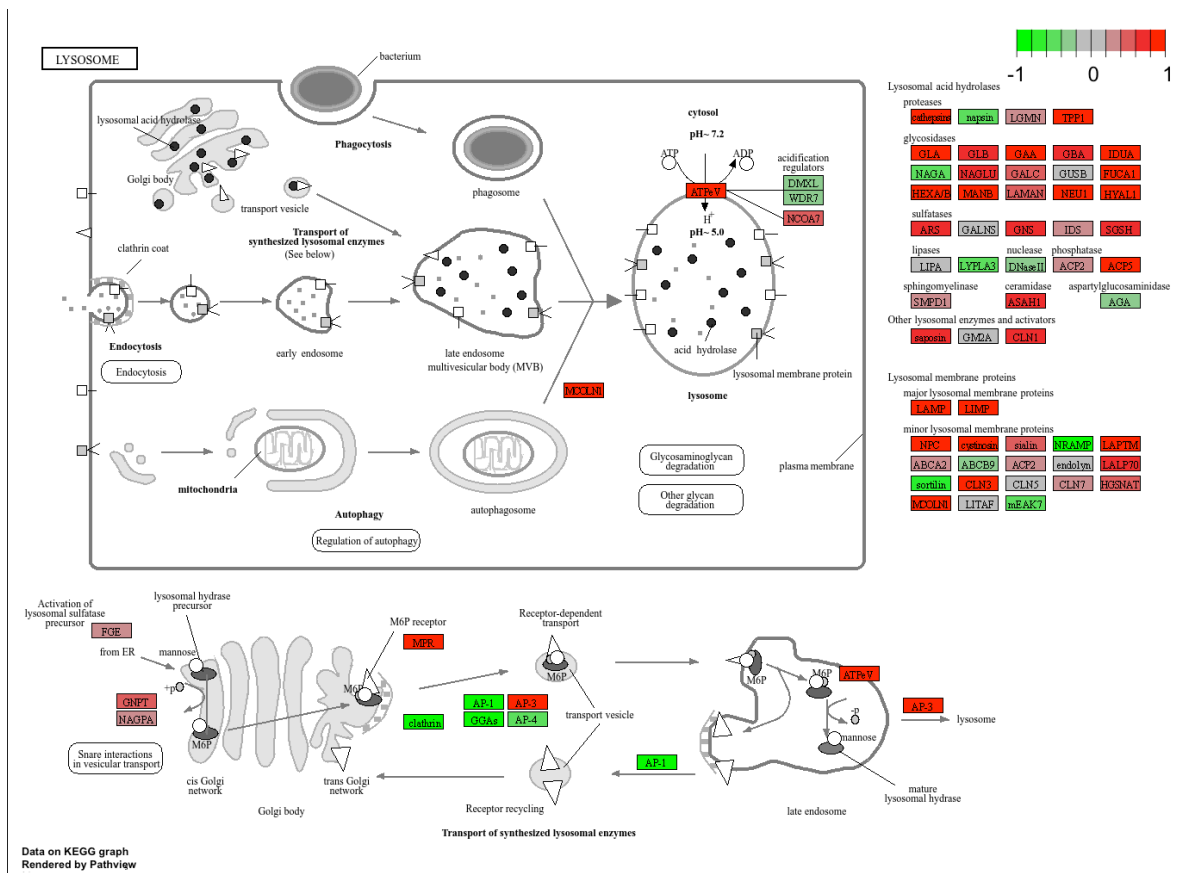


Figure 1: Hematopoietic cell lineage



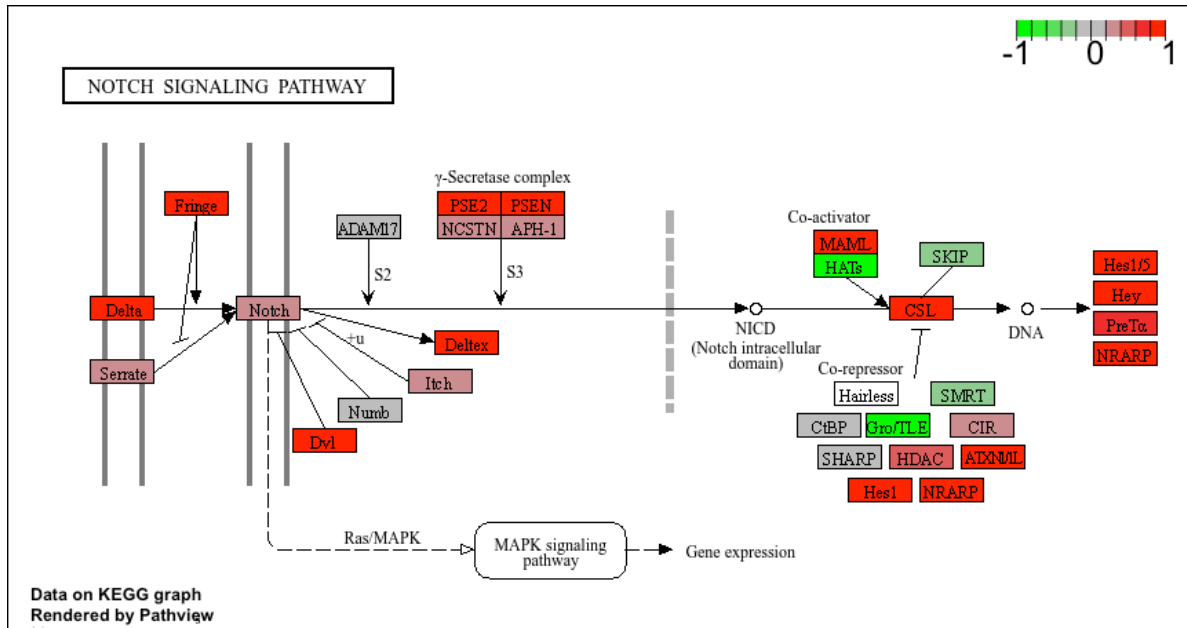


Figure 4: Lysosome

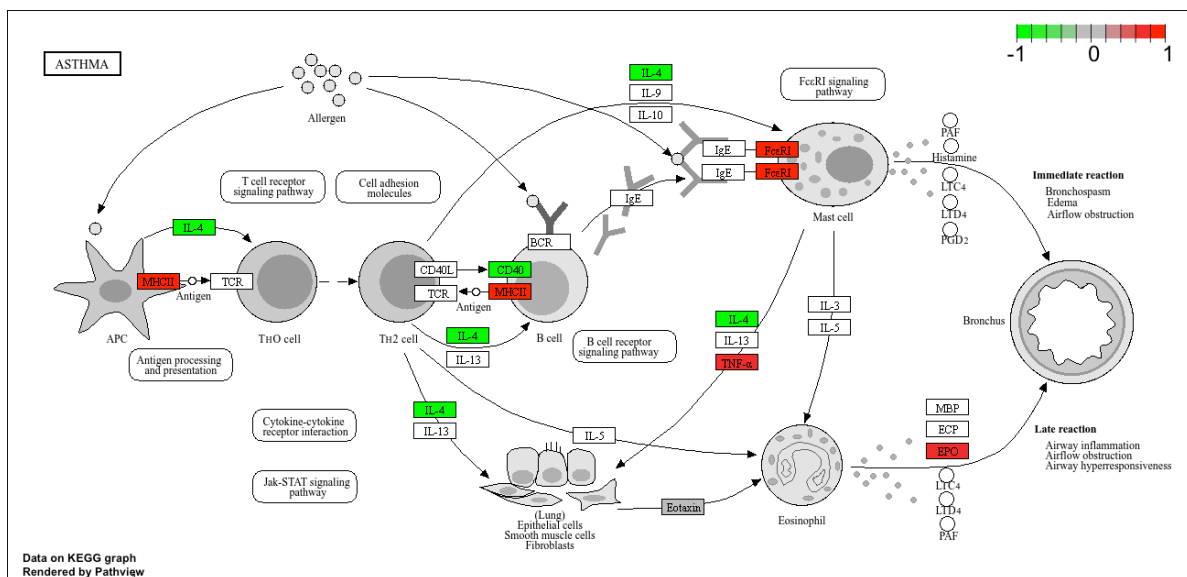


Figure 5: Notch signaling pathway

Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways_less <- rownames(keggres$less)[1:5]
```

```
# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways_less, start=1, stop=8)
keggresids
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/elizabethgilmore/Desktop/UCSD_Salk/BGGN213/BGGN213_github/
```

```
Info: Writing image file hsa04110.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/elizabethgilmore/Desktop/UCSD_Salk/BGGN213/BGGN213_github/
```

```
Info: Writing image file hsa03030.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/elizabethgilmore/Desktop/UCSD_Salk/BGGN213/BGGN213_github/
```

```
Info: Writing image file hsa03013.pathview.png
```

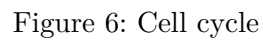
```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/elizabethgilmore/Desktop/UCSD_Salk/BGGN213/BGGN213_github/
```

```
Info: Writing image file hsa03440.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

Info: Writing image file hsa04114.pathview.png



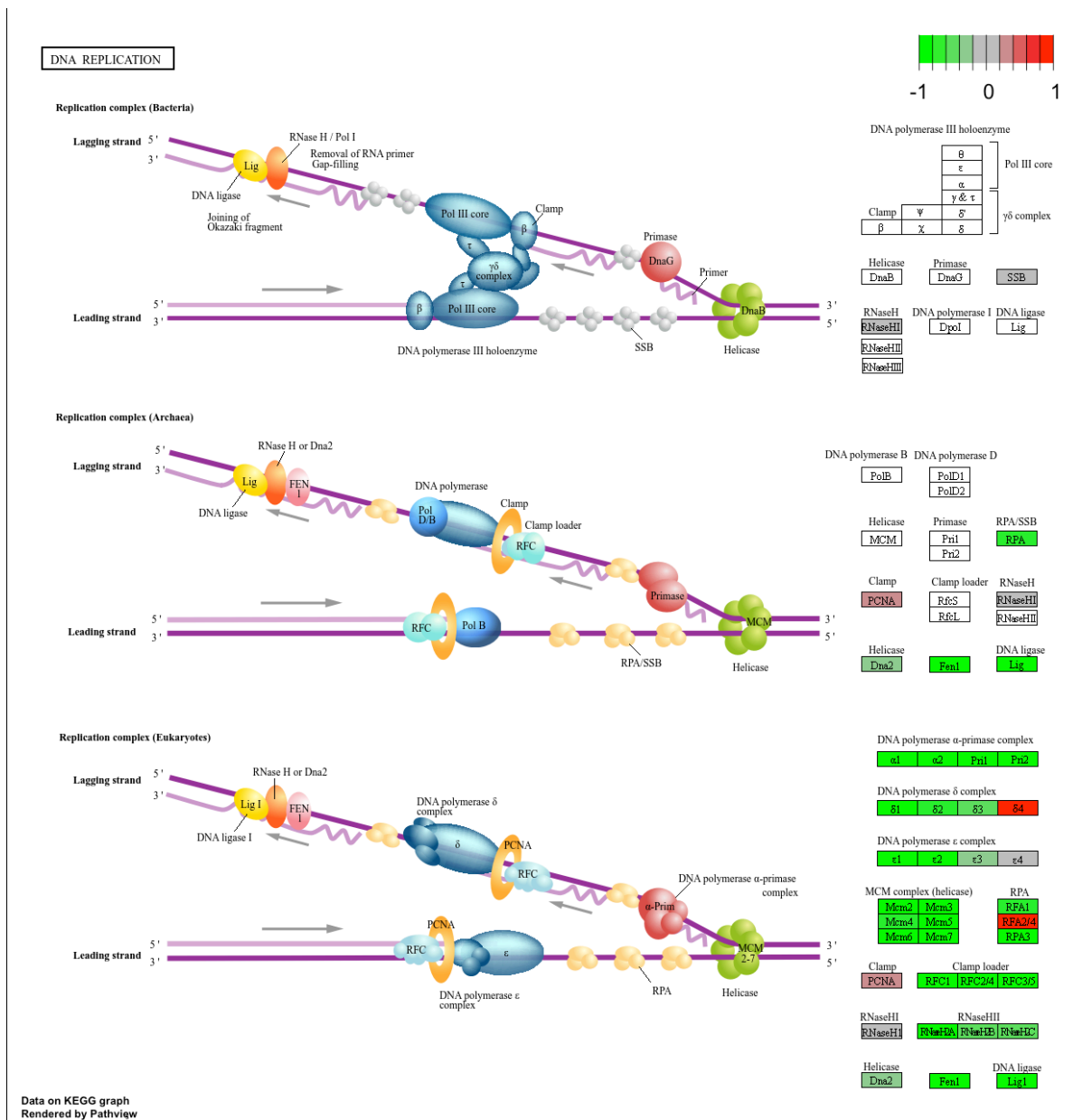


Figure 7: DNA replication

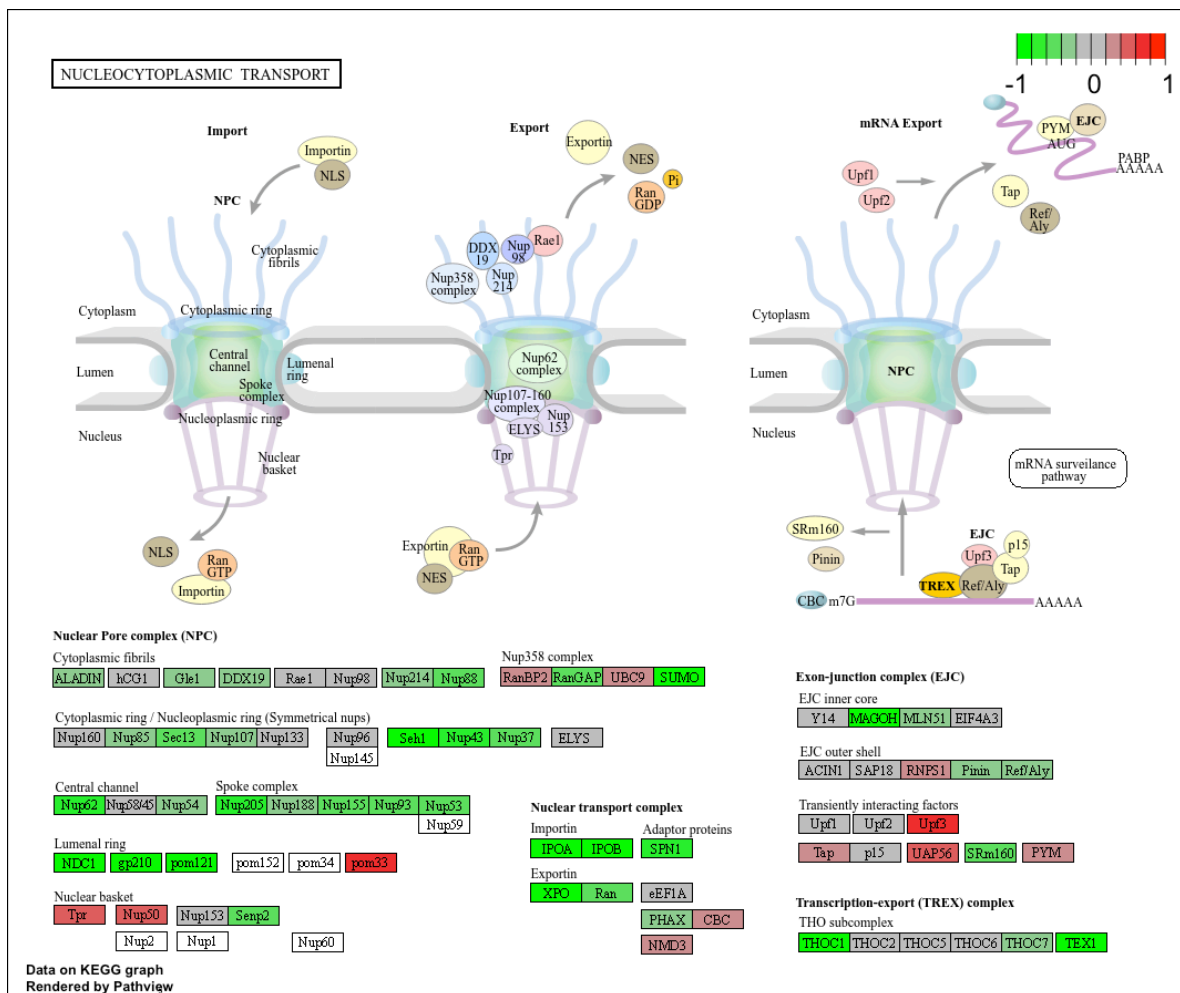


Figure 8: RNA transport


```

data(go.sets.hs)
data(go.subs.hs)

# Focus on biological process subset of G0
gobpsets <- go.sets.hs[go.subs.hs$BP]

gobpres <- gage(foldchanges, gsets=gobpsets)

lapply(gobpres, head)

```

\$greater

	p.geomean	stat.mean	p.val
G0:0007156 homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05
G0:0002009 morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
G0:0048729 tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
G0:0007610 behavior	1.925222e-04	3.565432	1.925222e-04
G0:0060562 epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
G0:0035295 tube development	5.953254e-04	3.253665	5.953254e-04
	q.val	set.size	expl
G0:0007156 homophilic cell adhesion	0.1951953	113	8.519724e-05
G0:0002009 morphogenesis of an epithelium	0.1951953	339	1.396681e-04
G0:0048729 tissue morphogenesis	0.1951953	424	1.432451e-04
G0:0007610 behavior	0.1967577	426	1.925222e-04
G0:0060562 epithelial tube morphogenesis	0.3565320	257	5.932837e-04
G0:0035295 tube development	0.3565320	391	5.953254e-04

\$less

	p.geomean	stat.mean	p.val
G0:0048285 organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280 nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067 mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059 chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236 mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10
	q.val	set.size	expl
G0:0048285 organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280 nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067 mitosis	5.841698e-12	352	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059 chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236 mitotic prometaphase	1.178402e-07	84	1.729553e-10

```
$stats
```

	stat.mean	exp1
G0:0007156 homophilic cell adhesion	3.824205	3.824205
G0:0002009 morphogenesis of an epithelium	3.653886	3.653886
G0:0048729 tissue morphogenesis	3.643242	3.643242
G0:0007610 behavior	3.565432	3.565432
G0:0060562 epithelial tube morphogenesis	3.261376	3.261376
G0:0035295 tube development	3.253665	3.253665

```
head(gobpres$less)
```

	p.geomean	stat.mean	p.val
G0:0048285 organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280 nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067 mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059 chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236 mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10

	q.val	set.size	exp1
G0:0048285 organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280 nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067 mitosis	5.841698e-12	352	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059 chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236 mitotic prometaphase	1.178402e-07	84	1.729553e-10

Reactome

Some folks really like reactome online rather than the R package of the same name (available from bioconductor).

To use the website viewer we want to upload our set of gene symbols for the genes we want to focus on (here those with a P vlaue below 0.05)

Then, to perform pathway analysis online go to the Reactome website (<https://reactome.org/PathwayBrowser/#>) Select “choose file” to upload your significant gene list. Then, select the parameters “Project to Humans”, then click “Analyze”.

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The mitosis pathway has the most significant Entities p-value. This matches our previous KEGG results, where the cell cycle pathway was the most significantly downregulated pathway. Both pathways are related to cell division and proliferation. Differences between the two methods could arise from variations in pathway definitions, gene annotations, and statistical approaches used in the analyses.

